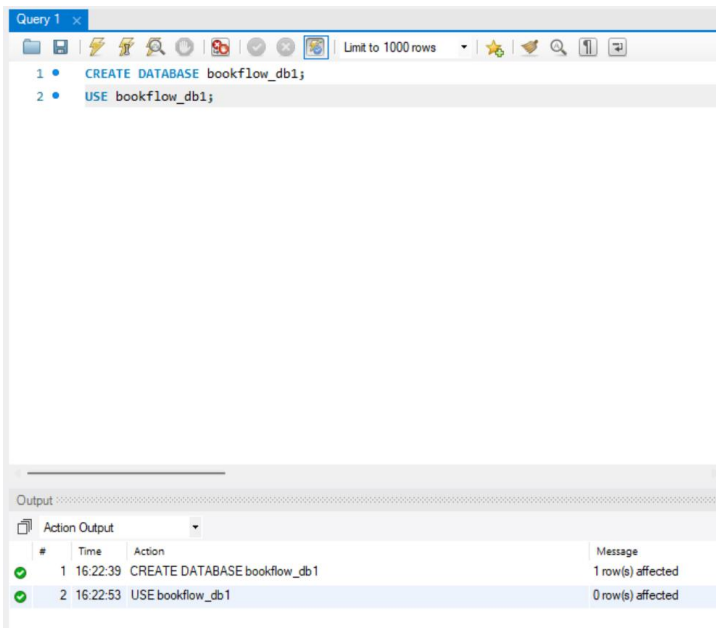


WEEK-2

MANASVI GHAG

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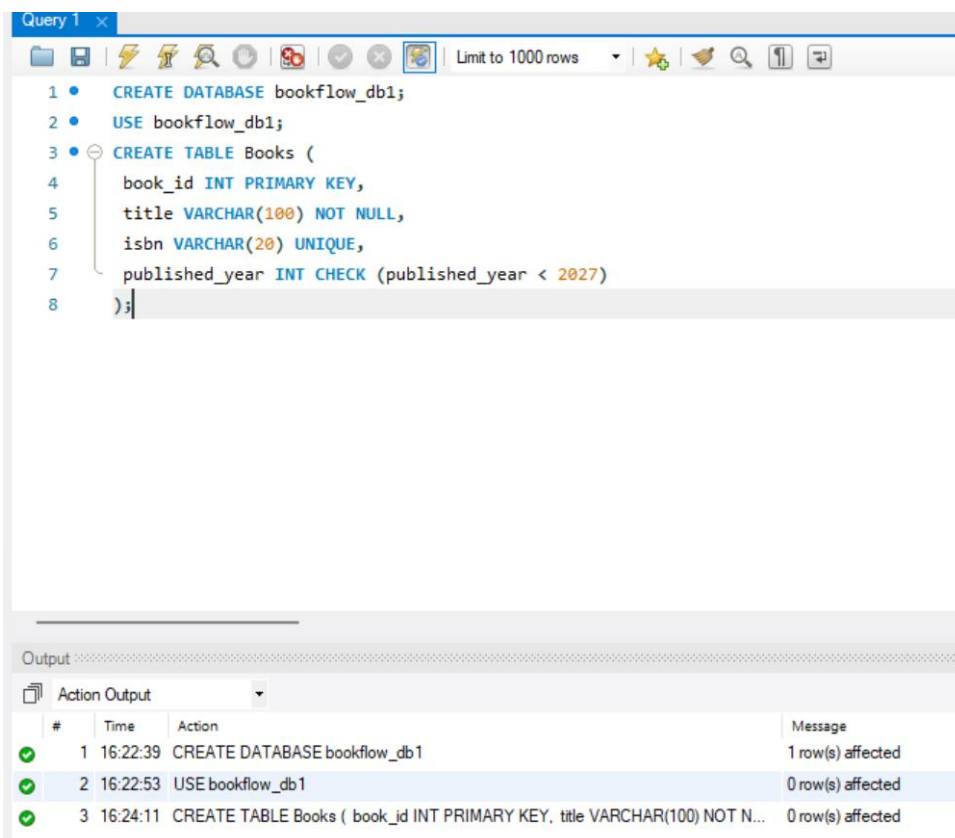
Step 1 — Create Database



```
Query 1 x
1 • CREATE DATABASE bookflow_db1;
2 • USE bookflow_db1;
```

#	Time	Action	Message
1	16:22:39	CREATE DATABASE bookflow_db1	1 row(s) affected
2	16:22:53	USE bookflow_db1	0 row(s) affected

Step 2 — Create Books Table (DDL + Constraints)



```
Query 1 x
1 • CREATE DATABASE bookflow_db1;
2 • USE bookflow_db1;
3 • CREATE TABLE Books (
4     book_id INT PRIMARY KEY,
5     title VARCHAR(100) NOT NULL,
6     isbn VARCHAR(20) UNIQUE,
7     published_year INT CHECK (published_year < 2027)
8 );
```

#	Time	Action	Message
1	16:22:39	CREATE DATABASE bookflow_db1	1 row(s) affected
2	16:22:53	USE bookflow_db1	0 row(s) affected
3	16:24:11	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT N...	0 row(s) affected

Step 3 & 4 — Insert Books and Display

Query 1

```
8 );
9 • INSERT INTO Books (book_id, title, isbn, published_year) VALUES
10 (1, 'The Great Gatsby', '9780743273565', 1925),
11 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
12 (3, '1984', '9780451524935', 1949);
13 • SELECT * FROM Books;
```

Result Grid

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949
* NULL	NULL	NULL	NULL

Books 1

Output

Action Output

#	Time	Action	Message
✓ 1	16:22:39	CREATE DATABASE bookflow_db1	1 row(s) affected
✓ 2	16:22:53	USE bookflow_db1	0 row(s) affected
✓ 3	16:24:11	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT N...	0 row(s) affected
✓ 4	16:25:05	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great Ga...	3 row(s) affected Records: 3
✓ 5	16:25:10	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned

Step 5, 6 & 7 — Members Table: Create, Insert, Display

Query 1

```
19 );
20 • INSERT INTO Members (member_id, full_name, email) VALUES
21 (101, 'Manasvi', 'manasvi@gmail.com'),
22 (102, 'Aswitha', 'aswitha@gmail.com'),
23 (103, 'tejaswini', 'tejaswini@gmail.com');
24 • SELECT * FROM Members;
```

Result Grid

member_id	full_name	email
101	Manasvi	manasvi@gmail.com
102	Aswitha	aswitha@gmail.com
103	tejaswini	tejaswini@gmail.com
* NULL	NULL	NULL

Members 2

Output

Action Output

#	Time	Action	Message
✓ 3	16:24:11	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT ...	0 row(s) affected
✓ 4	16:25:05	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great ...	3 row(s) affected Records: 3
✓ 5	16:25:10	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
✓ 6	16:26:04	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHA...	0 row(s) affected
✓ 7	16:27:08	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'Manasvi', 'm...	3 row(s) affected Records: 3
✓ 8	16:27:20	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned

Step 8 — Create Loans Table (FOREIGN KEYS)

```
26 • CREATE TABLE Loans (  
27     loan_id INT PRIMARY KEY,  
28     member_id INT,  
29     book_id INT,  
30     loan_date DATE,  
31     FOREIGN KEY (member_id) REFERENCES Members(member_id),  
32     FOREIGN KEY (book_id) REFERENCES Books(book_id)  
33 );
```

Output

Action Output

#	Time	Action	Message
4	16:25:05	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
5	16:25:10	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
6	16:26:04	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHA...	0 row(s) affected
7	16:27:08	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'Manasvi', 'm...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
8	16:27:20	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
9	16:28:40	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id I...	0 row(s) affected

Step 9 & 10 — Insert 10 Loans and Display

```
41 • SELECT * FROM Loans;  
42
```

Result Grid

	loan_id	member_id	book_id	loan_date
▶	1	101	1	2025-01-05
	2	102	2	2025-01-08
	3	103	3	2025-01-10
	4	101	2	2025-02-01
	5	102	1	2025-02-05
	6	103	2	2025-02-12
	7	101	3	2025-03-01
	8	102	3	2025-03-07
	9	103	1	2025-03-15

Result Grid
Form Editor

Step 11 — JOIN Query

```
43 • SELECT m.full_name AS Member_Name, b.title AS Book_Title  
44 FROM Loans l  
45 INNER JOIN Members m ON l.member_id = m.member_id  
46 INNER JOIN Books b ON l.book_id = b.book_id;  
47
```

Result Grid

	Member_Name	Book_Title
▶	Manasvi	The Great Gatsby
	Manasvi	To Kill a Mockingbird
	Manasvi	1984
	Manasvi	The Great Gatsby
	Aswitha	To Kill a Mockingbird
	Aswitha	The Great Gatsby
	Aswitha	1984
	tejaswini	1984
	tejaswini	To Kill a Mockingbird
	tejaswini	The Great Gatsby

Result Grid
Form Editor
Field Types

Step 12 — GROUP BY Query

```
4 /
48 • SELECT published_year, COUNT(book_id) AS Total_Books
49 FROM Books
50 GROUP BY published_year
51 ORDER BY published_year;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	published_year	Total_Books
▶	1925	1
	1949	1
	1960	1

Result Grid
Form Editor

Step 13–16 — Donation via TRANSACTION

```
53 • CREATE TABLE Donation_History (
54     donation_id INT PRIMARY KEY,
55     book_id INT,
56     donor_name VARCHAR(100),
57     donation_date DATE,
58     FOREIGN KEY (book_id) REFERENCES Books(book_id)
59 );
60 • START TRANSACTION;
61 • INSERT INTO Books (book_id, title, isbn, published_year)
62     VALUES (4, 'Animal Farm', '9780451526342', 1945);
63 • INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
64     VALUES (1, 4, 'Sahasra', CURDATE());
65 • COMMIT;
```

Output:

Action Output

#	Time	Action	Message
✓ 12	16:32:04	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER...	10 row(s) returned
✓ 13	16:33:29	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP...	3 row(s) returned
✓ 14	16:35:46	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id IN...	0 row(s) affected
✓ 15	16:35:50	START TRANSACTION	0 row(s) affected
✓ 16	16:36:00	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (4, 'Animal Far...	1 row(s) affected
✓ 17	16:36:14	INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_dat...	1 row(s) affected
✓ 18	16:36:18	COMMIT	0 row(s) affected

Step 17 & 18 — Index on ISBN + Fast Search

```
67 • CREATE INDEX idx_books_isbn ON Books(isbn);
68 • SELECT * FROM Books WHERE isbn = '9780451524935';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell C

	book_id	title	isbn	published_year
▶	3	1984	9780451524935	1949
*	NULL	NULL	NULL	NULL

Result Grid

Form Editor

Books 6 x

Apply

Revert

Context Help

Snippets

Output

Action Output

	#	Time	Action	Message
✓	14	16:35:46	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id IN...	0 row(s) affected
✓	15	16:35:50	START TRANSACTION	0 row(s) affected
✓	16	16:36:00	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (4, 'Animal Far...	1 row(s) affected
✓	17	16:36:14	INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_dat...	1 row(s) affected
✓	18	16:36:18	COMMIT	0 row(s) affected
✓	19	16:37:26	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓	20	16:37:29	SELECT * FROM Books WHERE isbn = '9780451524935' LIMIT 0, 1000	1 row(s) returned